ORIGINAL ARTICLE

Proteomics Signatures of Chronic Kidney Disease in Hypertension

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BACKGROUND: Hypertension is a major modifiable risk factor for chronic kidney disease (CKD), yet predicting which hypertensive individuals will progress to CKD remains challenging. We aimed to develop and validate a plasma protein-based risk score for incident CKD in hypertension and to identify causal proteins and potential therapeutic targets.

METHODS: Using data from the UK Biobank Pharma Proteomics Project, we included 22 258 hypertensive participants without baseline CKD. A protein risk score was developed in a training set (n=11 671) and validated in internal testing (n=7778) and external validation (Scotland/Wales, n=2809) cohorts. Hypothesis-generating Mendelian randomization using *cis*- protein quantitative trait loci was performed to prioritize proteins for functional follow-up.

RESULTS: A 13-protein risk score (C-index ≈0.78) was developed, with a simplified 3-protein panel (RNASE1 [pancreatic ribonuclease], IGFBP4 [insulin-like growth factor-binding protein 4], GDF15 [growth differentiation factor 15]) capturing most of its predictive power. Adding the 13-protein score to the clinical model significantly improved the C-index by 0.019 (95% CI, 0.011–0.026) in internal testing and 0.047 (95% CI, 0.013–0.090) in external validation cohorts, with significant net reclassification improvement and integrated discrimination improvement, while an estimated glomerular filtration rate-based polygenic risk score provided no meaningful improvement. Mendelian randomization analysis identified 7 proteins (BMPER [BMP-binding endothelial regulator protein], GFRA1 [GDNF family receptor alpha-1], CST3 [cystatin-C], CXCL16 [C-X-C motif chemokine 16], FOLR1 [folate receptor alpha], AMBP [protein AMBP], and STC1 [stanniocalcin-1]) with nominally significant associations with hypertensive CKD, all with higher levels increasing risk. Enrichment analyses highlighted protease inhibition and folate metabolism pathways. FOLR1 and STC1 emerged as druggable targets.

CONCLUSIONS: This study establishes a robust plasma protein signature for predicting hypertensive CKD and provides genetic evidence supporting several proteins as prioritized candidates, revealing novel pathways and nominating FOLR1 and STC1 as potential therapeutic targets for further investigation. ***(Hypertension.* 2026;83:2252–2261. DOI: 10.1161/HYPERTENSIONAHA.125.26477.)** • **Supplement Material**.

Key Words: biomarkers ■ hypertension ■ proteomics ■ renal insufficiency, chronic ■ risk assessment

Hypertension affects over 1.3 billion adults worldwide and is the second leading modifiable risk factor for chronic kidney disease (CKD).[1,2] Despite effective blood pressure control, progression to CKD remains difficult to predict in individual patients, underscoring the need for improved risk stratification and mechanistic insight in this high-risk population.

Plasma proteins integrate genetic, environmental, and clinical influences, offering dynamic pathways of disease.[3] Large-scale proteomic platforms enable simultaneous measurement of thousands of proteins, facilitating biomarker discovery and support causal inference via Mendelian randomization (MR), aiding in drug target identification.[4] Protein-based risk scores have shown promise in related settings, for example, improving CKD prediction in diabetes,[5] but their utility in hypertension, and their added value over clinical models and polygenic risk scores (PRSs), remain unexplored.

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NOVELTY AND RELEVANCE

What Is New?

Using large-scale plasma proteomics in 22 258 hypertensive individuals, we developed and validated a 13-protein risk score that significantly improves incident chronic kidney disease prediction beyond established clinical models.

A simplified 3-protein panel (RNASE1 [pancreatic ribonuclease], IGFBP4 [insulin-like growth factor-binding protein 4], GDF15 [growth differentiation factor 15]) captures 98% to 99% of the full score's predictive performance, offering potential for clinical translation.

Hypothesis-generating Mendelian randomization analysis prioritized 7 proteins (BMPER [BMP-binding endothelial regulator protein], GFRA1 [GDNF family receptor alpha-1], CST3 [cystatin-C], CXCL16 [C-X-C motif chemokine 16], FOLR1 [folate receptor alpha], AMBP [protein AMBP], and STC1 [stanniocalcin-1]) with causal relevance to hypertensive chronic kidney disease, implicating protease regulation and folate metabolism pathways.

What Is Relevant?

While protein risk scores have improved chronic kidney disease prediction in diabetes, their utility in hypertension and their added value over clinical models and polygenic risk scores remained unexplored. This study fills that gap.

Clinical/Pathophysiological Implications?

The protein risk score, particularly the 3-protein panel, could enable identification of high-risk hypertensive patients for intensified monitoring and early intervention (eg, stricter blood pressure control, sodium-glucose cotransporter 2 inhibitors).

The findings shift the paradigm from traditional risk factors to a pathway-based understanding of chronic kidney disease risk in hypertension.

The integrated Mendelian randomization-druggability pipeline nominates FOLR1 and STC1 as potential therapeutic targets, with FOLR1 representing a particularly promising candidate for further investigation.

Nonstandard Abbreviations and Acronyms

AMBP	protein AMBP
BMPER	BMP-binding endothelial regulator protein
CKD	chronic kidney disease
CKD-PC	Chronic Kidney Disease Prognosis Consortium
CST3	cystatin-C
CXCL16	C-X-C motif chemokine 16
eGFR	estimated glomerular filtration rate
FOLR1	folate receptor alpha
GCKR	glucokinase regulatory protein
GDF15	growth differentiation factor 15
GFRA1	GDNF family receptor alpha-1
IGFBP4	insulin-like growth factor-binding protein 4
MR	Mendelian randomization
NRG4	neuregulin-4
pQTL	protein quantitative trait loci
PRS	polygenic risk score
RNASE1	pancreatic ribonuclease
STC1	stanniocalcin-1
TCEA2	transcription elongation factor A protein 2

Here, we leveraged data from the UK Biobank Pharma Proteomics Project, which measured ≈3000 plasma proteins in ≈50 000 participants, to address 2 key questions: (1) whether a protein risk score can improve prediction of incident CKD in hypertensive individuals beyond the established CKD Prognosis Consortium (CKD-PC) clinical model[6] and an estimated glomerular filtration rate (eGFR)-based PRS; and (2) whether MR using *cis*-pQTLs (*cis*-acting protein quantitative trait loci) as instruments can identify proteins with causal effects on hypertensive CKD, thereby nominating novel pathways and potential therapeutic targets.

METHODS

Data Availability

The UK Biobank data are available on application to the UK Biobank (https://www.ukbiobank.ac.uk/).

Study Design and Participants

The UK Biobank recruited over 500 000 adults aged 40 to 69 years across the UK between 2006 and 2010.[7–9] The UK Biobank Pharma Proteomics Project measured ≈3000 plasma proteins in a random subset of ≈50 000 participants.[10–12] All participants provided written informed consent, and the study was approved by the North West Research Ethics Committee (11/NW/0382).

From the UK Biobank Pharma Proteomics Project, we identified 29 906 participants with hypertension, defined as systolic blood pressure ≥140 mm Hg, diastolic blood pressure ≥90 mm Hg, use of antihypertensive medication, or a documented diagnosis of hypertension. After excluding individuals with withdrawn consent (n=7), missing CKD status (n=2237), prevalent CKD at baseline (n=3776; defined as eGFR <60 mL/min per 1.73 m^2, urinary albumin-to-creatinine ratio ≥30 mg/g, or self-reported/hospital-diagnosed CKD), missing genetic data or sex mismatch (n=117), or missing covariates for the CKD-PC clinical model (n=1511), 22 258 participants were included for analysis. Baseline characteristics were comparable between these participants and the 186 777 hypertensive individuals without proteomic data in UK Biobank, suggesting no substantial selection bias (Table S1).

Participants were divided into a discovery cohort (England, n=19 449) and an independent validation cohort (Scotland/Wales, n=2809). The discovery cohort was randomly split into training (60%, n=11 671) and testing (40%, n=7778) sets. A participant flowchart is provided in Figure S1. The study design is shown in Figure S2.

Clinical Risk Factors and Measurements

We constructed a clinical prediction model based on the established CKD-PC model,[6] including age, sex, ethnicity, smoking status, systolic blood pressure, body mass index, eGFR, urinary albumin-to-creatinine ratio, and history of cardiovascular disease and diabetes.

Baseline characteristics and medical history were collected via touchscreen questionnaires. Blood pressure was measured twice using a digital sphygmomanometer (Omron 705 IT) after at least 5 minutes of seated rest; the average of the 2 readings was used. Body mass index was calculated as weight (kg) divided by height squared (m^2).

Serum creatinine was measured enzymatically (Beckman Coulter AU5800), and eGFR was calculated using the Chronic Kidney Disease Epidemiology Collaboration (CKD-EPI) equation.[13] Urinary albumin and creatinine were measured on a Beckman Coulter AU5400 system via immunoturbidimetric and enzymatic methods, respectively, from which the urinary albumin-to-creatinine ratio was derived. Glycated hemoglobin was measured by high-performance liquid chromatography (Bio-Rad Variant II Turbo). Cardiovascular disease was defined as self-reported or medically recorded coronary heart disease, stroke, or heart failure. Diabetes was defined as a documented diagnosis, use of glucose-lowering medication, or glycated hemoglobin ≥48 mmol/mol.[14–16]

Proteomic Assessment

Plasma proteomic profiling of UK Biobank baseline samples was conducted by the UK Biobank and Olink facilities using the Olink Explore 3072 proximity extension assay, as previously described in detail.[10,17] The platform quantified 2941 protein analytes, corresponding to 2923 unique proteins. Sample handling, randomization, sequencing (NovaSeq 6000), and initial quality control procedures were conducted by the UK Biobank and Olink facilities, with normalization performed by Olink to produce inverse rank-based normalized protein expression values on a $\log_2$ scale. Our study used these preprocessed data from the baseline assessment. For the present analysis, we excluded 12 proteins with >20% missingness, leaving 2911 proteins for analysis. Missing values in the remaining proteins were imputed with the mean, and each protein was standardized by dividing by its SD for analysis as a continuous variable.

Assessment of Outcome

The primary outcome was incident CKD, identified using the validated first-occurrence algorithm in UK Biobank, which defines CKD based on *International Classification of Diseases*, Tenth Revision code N18 from linked primary care, hospital admission, and death registry data (https://biobank.ndph.ox.ac.uk/ukb/label.cgi?id=1712).[18] Follow-up time accrued from the date of recruitment to the first diagnosis of CKD, death, loss to follow-up, or the end of the follow-up period, whichever occurred first.

In sensitivity analyses, incident CKD was also defined using *International Classification of Diseases*, Ninth Revision codes 585 and 5859, *International Classification of Diseases*, Tenth Revision codes I12.0, I13.1, I13.2, N18.0, N18.3, N18.4, N18.5, N18.8, N18.9, and Office of Population Censuses and Surveys Classification of Interventions and Procedures-4 code M01.[9,19]

eGFR-Based PRS

We constructed a weighted eGFR-based PRS using 263 single nucleotide polymorphisms independently associated with eGFR at genome-wide significance in a genome-wide association studies (GWAS) meta-analysis excluding UK Biobank participants.[20,21] Genotyping, imputation, and quality control procedures for UK Biobank have been described previously.[22]

For each individual, the PRS was calculated as the sum of eGFR-increasing alleles, weighted by the corresponding regression coefficients from the discovery GWAS.[21,23,24] Each single nucleotide polymorphism was coded as 0, 1, or 2 according to the number of eGFR-increasing alleles. A higher PRS indicates higher genetically predicted eGFR and thus lower genetic susceptibility to CKD. The full list of 263 single nucleotide polymorphisms is provided in Table S2.

Statistical Analyses

Descriptive Analysis

Baseline characteristics were summarized as means (SD) for continuous variables and counts (%) for categorical variables. Differences across training, testing, and validation sets were assessed using ANOVA and χ^2 tests, as appropriate.

Protein Risk Score Development and Assessment

The protein risk score was developed in the training set (n=11 671; 740 events) and evaluated in the testing (n=7778; 492 events) and validation (n=2809; 72 events) sets. First, Cox regression adjusted for clinical covariates comprising the CKD-PC model identified 264 proteins associated with incident CKD at a Bonferroni-corrected threshold (P<0.05; Table S3). These candidates, together with age and sex, were included in a least absolute shrinkage and selection operator Cox model with 10-fold cross-validation to select the optimal predictor subset (Figure S3). The final protein risk score was calculated as the weighted sum of selected proteins, with coefficients reflecting effect direction (positive for risk, negative for protective).

In the testing and validation sets, Cox models estimated hazard ratios and 95% CIs for the association between the protein risk score and incident CKD, adjusting for clinical risk factors from the CKD-PC model and the eGFR-based PRS. Model performance was assessed by discrimination (Harrell C-index) and reclassification (10-year continuous net reclassification improvement and integrated discrimination improvement), with 95% CIs estimated by bootstrapping (1000 replicates).

Subgroup analyses were conducted by prior medical diagnosis of hypertension (yes versus no) and follow-up duration. For the latter, we stratified incident CKD events into tertiles according to the time interval between blood collection and diagnosis, and compared model performance (C-index) for events occurring in the first tertile (shortest follow-up) versus the combined second and third tertiles (longer follow-up). Density plots were generated to visualize the distribution of time-to-event across cohorts.

MR, Functional Analysis, and Druggable Protein Identification

As a secondary, hypothesis-generating analysis, we performed 2-sample MR using the R package TwoSampleMR to prioritize proteins for functional follow-up by assessing potential causal effects on hypertensive CKD. Genetic instruments were *cis*-pQTLs from the UK Biobank Pharma Proteomics Project (n=34 557), selected as independent variants ($P<5\times10^{-8}$; LD $r^2<0.01$ within a 5000 kb window, located within ±500 kb of the transcription start site),[25] satisfying the relevance assumption that the instruments are strongly associated with protein levels. Summary-level data for hypertensive renal disease were obtained from the FinnGen consortium (R11 release, I9_HYPTENSRD), including 1217 cases and 316 345 controls of European ancestry.[26] Importantly, the reference population was the general population without CKD, not restricted to individuals with hypertension, which minimizes the risk of collider bias and supports the independence assumption by reducing selection bias. In total, 2114 *cis*-pQTLs corresponding to 264 candidate proteins were analyzed. For proteins with a single pQTL, the Wald ratio was used; for multiple pQTLs, inverse-variance weighted meta-analysis was applied.[27] Associations with $P<0.05$ were considered significant.

Colocalization analysis using the R package coloc was performed to assess whether the associations between the identified proteins and for hypertensive kidney disease resulted from linkage disequilibrium.[28] For each protein, we computed posterior probabilities for 5 exclusive hypotheses, with priors set at $p1=1\times10^{-4}$ (trait 1 only), $p2=1\times10^{-4}$ (trait 2 only), and $p12=1\times10^{-5}$ (shared variant). Strong colocalization was defined as posterior probability for a shared causal variant (PH4)≥0.8; moderate colocalization as 0.5< PH4<0.8.

Proteins significant in both Cox and MR analyses were tested for enrichment in Gene Ontology and Kyoto Encyclopedia of Genes and Genomes pathways using the R package clusterProfiler,[29] with significance assessed by Fisher exact test with false discovery rate correction. Druggability was evaluated by querying the Therapeutic Target Database.

All analyses were performed in R (version 4.1.1). A 2-tailed $P<0.05$ was considered statistically significant unless otherwise specified.

RESULTS

Study Population Characteristics

The discovery cohort comprised 19 449 hypertensive participants (mean age, 58.8 years [SD, 7.5]; 46.1% men). During a median follow-up of 13.4 years (interquartile range, 12.7–14.1), 1232 incident CKD cases (6.3%) were documented, with comparable event rates in the training (n=11 671; 740 events, 6.3%) and internal testing (n=7778; 492 events, 6.3%) subsets. Baseline characteristics were well balanced between the 2 groups (Table 1). As shown in Figure S4, incident CKD events occurred throughout the follow-up period in both cohorts. While some temporal variation in event intensity was observed, notably an early peak around 3 to 4 years in the discovery cohort, events continued to accrue steadily over the entire study period. This distribution, with no evidence of exclusive early clustering, supports the suitability of the protein risk score for long-term risk stratification.

The external validation cohort included 2809 hypertensive participants from Scotland and Wales. Over a longer median follow-up of 14.4 years (interquartile range, 14.1–14.6), 72 incident CKD cases (2.6%) were identified. Compared with the discovery cohort, participants in the validation cohort were more likely to be non-Black and never smokers, had higher systolic and diastolic blood pressure at baseline, and reported lower use of antihypertensive or glucose-lowering medications. These differences, together with the lower observed CKD incidence, likely reflect regional variations in demographic characteristics and health care practices, and underscore the importance of external validation in a distinct population (Table 1).

Candidate Protein Profiling, Protein Risk Score, and Incident CKD

Of 2911 plasma proteins analyzed in the training set, 264 were significantly associated with incident CKD after Bonferroni correction ($P<0.05$; Table S3). Least absolute shrinkage and selection operator regression selected 13 proteins from these candidates for the final risk score, all positively associated with CKD risk (Table S4). The 3 proteins with the largest absolute coefficients, thus the greatest contribution to the score, were RNASE1 (pancreatic ribonuclease), IGFBP4 (insulin-like growth factor-binding protein 4), and GDF15 (growth differentiation factor 15).

After adjustment for clinical factors from the CKD-PC model and the eGFR-based PRS, each SD increase in the protein risk score remained significantly associated with higher CKD risk in both the testing set (adjusted hazard ratio, 1.67 [95% CI, 1.53–1.83]) and the validation cohort (adjusted hazard ratio, 2.33 [95% CI, 1.84–2.94]; Figure 1).

Table 1. Baseline Characteristics of Study Participants*

Characteristics	Total participant†		Discovery cohort	
	Discovery cohort	Validation cohort	Training set	Testing set
N	19449	2809	11671	7778
Age, y	58.82 (7.51)	58.59 (7.44)	58.79 (7.47)	58.87 (7.59)
Male, n (%)	10201 (52.4)	1478 (52.6)	6152 (52.7)	4049 (52.1)
Black, n (%)	442 (2.3)	10 (0.4)	253 (2.2)	189 (2.4)
Body mass index, kg/m^2	28.35 (4.81)	28.59 (4.76)	28.35 (4.83)	28.35 (4.79)
Never smokers, n (%)	10221 (52.6)	1551 (55.2)	6158 (52.8)	4063 (52.2)
Estimated glomerular filtration rate, mL/min per 1.73 m^2	90.22 (11.83)	90.86 (11.55)	90.37 (11.75)	89.98 (11.96)
Urinary albumin-to-creatinine ratio, mg/g	9.35 (5.81)	9.15 (5.75)	9.32 (5.79)	9.41 (5.84)
Glycated hemoglobin, %	5.52 (0.63)	5.50 (0.62)	5.52 (0.63)	5.52 (0.64)
Systolic blood pressure, mm Hg	148.50 (15.78)	149.69 (15.20)	148.65 (15.67)	148.27 (15.93)
Diastolic blood pressure, mm Hg	86.90 (9.48)	87.78 (9.34)	86.95 (9.48)	86.81 (9.49)
Prevalent disease, n (%)				
Cardiovascular disease	2005 (10.3)	298 (10.6)	1205 (10.3)	800 (10.3)
Diabetes	1533 (7.9)	211 (7.5)	927 (7.9)	606 (7.8)
Diagnosed hypertension	10 129 (52.1)	1403 (49.9)	6029 (51.7)	4100 (52.7)
Medications, n (%)				
Antihypertensive drugs	7088 (36.4)	989 (35.2)	4161 (35.7)	2927 (37.6)
Glucose-lowering drugs	924 (4.8)	114 (4.1)	565 (4.8)	359 (4.6)
Incident chronic kidney disease, n (%)	1232 (6.3)	72 (2.6)	740 (6.3)	492 (6.3)

*Data are presented as mean (SD) for continuous variables or n (%) for categorical variables.
†The median follow-up was 12.8 y (interquartile range [IQR], 12.0–13.8) for the discovery cohort and 13.6 y (IQR, 12.7–14.3) for the validation cohort.

Discrimination and Reclassification Ability

In the testing set, all models improved upon the age-sex reference (C-index, 0.680 [95% CI, 0.657–0.702]). The protein risk score achieved a C-index of 0.787 (95% CI, 0.767–0.807), followed closely by the top 3-protein model (0.774 [95% CI, 0.754–0.794]), while the eGFR-based PRS added minimal gain (0.692 [95% CI, 0.670–0.715]). The clinical model alone yielded a C-index of 0.811 (95% CI, 0.792–0.829). A similar pattern was observed in the validation cohort: protein risk score (0.779 [95% CI, 0.720–0.837]), top 3-protein model (0.773 [95% CI, 0.716–0.831]), and clinical model (0.760 [95% CI, 0.701–0.819]) all substantially outperformed the age-sex reference (0.644 [95% CI, 0.583–0.706]), whereas the eGFR-based PRS again provided negligible improvement (0.646 [95% CI, 0.584–0.707]; Table 2). Notably, the top 3 proteins captured most of the predictive capacity of the full score, achieving 98.3% and 99.2% of its performance in the testing and validation sets, respectively.

Adding the protein risk score to the clinical model further improved discrimination, with C-index increases of 0.019 (95% CI, 0.011–0.026) in the testing set and 0.047 (95% CI, 0.013–0.090) in the validation cohort. The top 3-protein model yielded similar gains (ΔC-index, 0.016 [95% CI, 0.009–0.023] in the testing set and 0.044 [95% CI, 0.011–0.084] in the validation cohort, respectively). In contrast, adding the eGFR-based PRS provided no meaningful improvement (ΔC-index, 0.001 [95% CI, −0.001 to 0.003] and 0.000 [95% CI, −0.006 to 0.004]; Table 2). When added to the clinical model, both protein-based models also significantly improved reclassification, with positive integrated discrimination improvement and net reclassification improvement values across cohorts, while the eGFR-based PRS showed negligible effects (Table 2). Results were consistent in sensitivity analyses using alternative CKD definitions (Table S5) and across subgroups stratified by prior hypertension diagnosis (Table S6). In subgroup analyses stratified by follow-up duration, all models demonstrated robust performance for both short-term (first tertile) and longer-term (combined second and third tertiles) events. Notably, C-indices were slightly higher for longer-term events (Table S7), suggesting that the proteomic risk score maintains, even potentially improves, its predictive accuracy over longer time horizons.

MR Analyses

We performed 2-sample MR using 2114 *cis*-pQTLs corresponding to the 264 Cox-derived candidate proteins to assess potential causal effects on hypertensive CKD. At a nominal significance threshold (P<0.05), inverse-variance weighted analysis identified 7 proteins for which genetically predicted higher levels were associated with increased CKD risk: BMP-binding endothelial regulator protein (BMPER), GFRA1 (GDNF family receptor alpha-1), CST3 (cystatin-C), CXCL16 (C-X-C motif

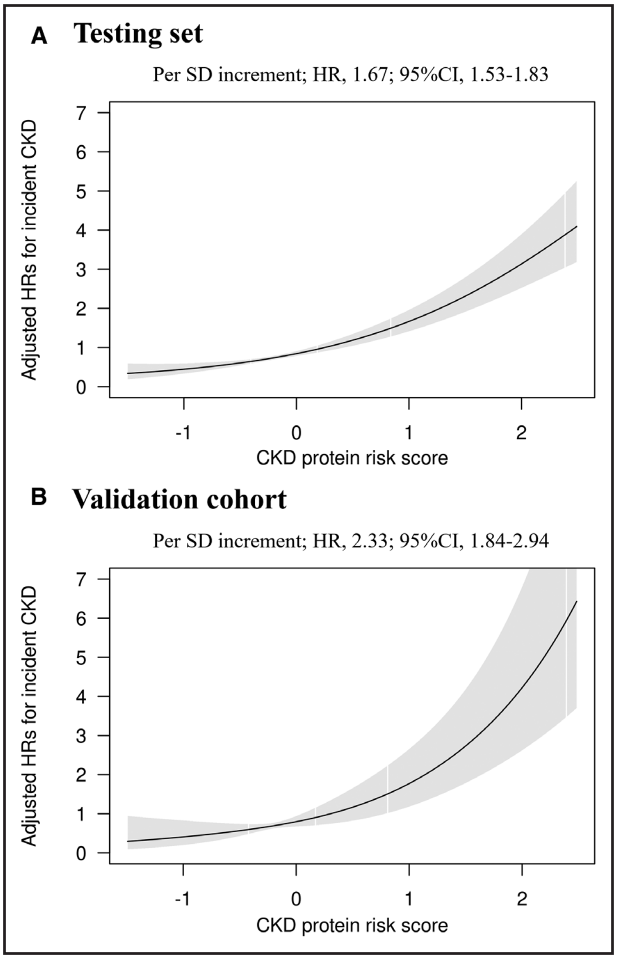

Figure 1. Dose-response association between protein risk score and incident chronic kidney disease (CKD).
A, Testing set. **B**, Validation cohort. Restricted cubic spline curves show adjusted hazard ratios (HRs) (solid lines) with 95% CIs (shaded) across the protein risk score continuum. Interpretation: Values above HR=1 indicate increased risk. The continuous upward trend in both cohorts suggests a graded positive association without threshold effects. Adjusted for age, sex, ethnicity, body mass index, smoking status, estimated glomerular filtration rate (eGFR), urinary albumin-to-creatinine ratio, systolic blood pressure, cardiovascular disease, diabetes, and eGFR-based polygenic risk score.

chemokine 16), FOLR1 (folate receptor alpha), AMBP (protein AMBP), and STC1 (stanniocalcin-1; Figure 2; Tables S3 and S8). Effect directions were consistent with those observed in Cox regression. However, none of these associations survived false discovery rate correction (Table S8). Colocalization analyses showed strong support for association with protein levels (PH1 range, 0.62–0.91) but limited evidence for a shared causal variant with CKD (PH4<0.5 for all; Table S9).

Enrichment Analyses of Causal Proteins

Molecular function enrichment analysis of the 7 MR-identified proteins revealed significant overrepresentation of protease regulation pathways, including endopeptidase regulator activity, peptidase regulator activity, and enzyme inhibitor activity. Kyoto Encyclopedia of Genes and Genomes pathway analysis highlighted folate transport and metabolism pathways and antifolate resistance (Table S10).

Druggability of Candidate Proteins

Among the 7 candidate proteins, FOLR1 and STC1 have documented therapeutic targeting. FOLR1 is targeted by 2 approved drugs (mirvetuximab soravtansine for ovarian cancer; pyrimethamine for malaria) and 10 additional agents in clinical trials, primarily for solid tumors, with mechanisms including antagonism and inhibition (Table S11). STC1 has been proposed as a potential target in breast cancer. Neither protein has been explored therapeutically in CKD.

DISCUSSION

This large-scale proteomic study demonstrates that plasma protein data significantly enhance the prediction of incident CKD and provide insights into its pathophysiology in hypertension. We developed a 13-protein risk score that substantially improved prediction beyond established clinical models, with a simplified 3-protein panel (RNASE1, IGFBP4, GDF15) capturing nearly identical performance. MR analysis identified 7 proteins with putative causal links to CKD, implicating pathways of protease regulation and folate metabolism. Druggability assessment nominated FOLR1 and STC1 as potential therapeutic targets, though not yet explored in CKD.

Proteomic Profiling Enhances CKD Risk Prediction in Hypertension

Adding the protein risk score to the clinical model significantly improved discrimination (ΔC-index 0.019 in the testing set; 0.047 in the validation cohort) and reclassification (positive net reclassification improvement and integrated discrimination improvement) across cohorts. These gains, while modest in absolute terms, have meaningful population-level implications for risk stratification: even a 1.9% improvement in discrimination could translate into more accurate risk classification for thousands of individuals when applied at scale. The larger improvement observed in the external cohort likely reflects differences in demographic composition, smaller sample size, and lower event rate, underscoring the need for validation in larger, more diverse populations. Together, these findings suggest that plasma proteins capture pathophysiological signals not reflected by conventional markers such as eGFR, urinary albumin-to-creatinine ratio, or blood pressure,[30] offering a dynamic window into active disease processes that may enhance early

Table 2. Predictive Performance of the Protein Risk Score for Incident Chronic Kidney Disease in the Testing Set and Validation Cohorts

Model	C-index (95% CI)	Comparison to reference models: C-index increase (95% CI)	10-y risk IDI (95% CI)	10-y risk overall cNRI (95% CI)
Testing set				
Age+sex	0.680 (0.657–0.702)			
Reference model: age+sex				
Age+sex+polygenic risk score	0.692 (0.670–0.715)	0.013 (0.003–0.022)	0.003 (0.001–0.006)	0.092 (0.025–0.148)
Age+sex+protein risk score	0.787 (0.767–0.807)	0.107 (0.086–0.128)	0.059 (0.042–0.080)	0.384 (0.321–0.438)
Age+sex+top 3 proteins†	0.774 (0.754–0.794)	0.094 (0.073–0.114)	0.035 (0.017–0.050)	0.343 (0.276–0.403)
Clinical risk factors*	0.811 (0.792–0.829)	0.131 (0.108–0.153)	0.090 (0.071–0.119)	0.465 (0.411–0.520)
Reference model: clinical risk factors*				
Clinical risk factors*+polygenic risk score	0.812 (0.793–0.830)	0.001 (−0.001 to 0.003)	0.000 (−0.001 to 0.003)	0.045 (−0.047 to 0.108)
Clinical risk factors*+protein risk score	0.829 (0.812–0.847)	0.019 (0.011–0.026)	0.022 (0.011–0.037)	0.143 (0.059–0.205)
Clinical risk factors*+top 3 proteins†	0.827 (0.809–0.844)	0.016 (0.009–0.023)	0.017 (0.006–0.029)	0.118 (0.040–0.198)
Validation cohort				
Age+sex	0.644 (0.583–0.706)			
Reference model: age+sex				
Age+sex+polygenic risk score	0.646 (0.584–0.707)	0.001 (−0.017 to 0.013)	0.001 (0.000–0.006)	0.087 (−0.085 to 0.234)
Age+sex+protein risk score	0.779 (0.720–0.837)	0.134 (0.071–0.202)	0.059 (0.025–0.139)	0.474 (0.274–0.612)
Age+sex+top 3 proteins†	0.773 (0.716–0.831)	0.129 (0.068–0.194)	0.060 (0.025–0.141)	0.445 (0.255–0.597)
Clinical risk factors*	0.760 (0.701–0.819)	0.116 (0.047–0.168)	0.027 (0.010–0.099)	0.304 (0.132–0.488)
Reference model: clinical risk factors*				
Clinical risk factors*+polygenic risk score	0.760 (0.702–0.819)	0.000 (−0.006 to 0.004)	0.000 (−0.001 to 0.009)	0.002 (−0.100 to 0.179)
Clinical risk factors*+protein risk score	0.807 (0.756–0.858)	0.047 (0.013–0.090)	0.059 (0.023–0.140)	0.305 (0.113–0.474)
Clinical risk factors*+top 3 proteins†	0.804 (0.753–0.855)	0.044 (0.011–0.084)	0.053 (0.017–0.134)	0.299 (0.076–0.450)

cNRI indicates continuous net reclassification index; and IDI, integrated discrimination improvement.

*Clinical risk factors included age, sex, ethnicity, body mass index, smoking status, estimated glomerular filtration rate, urinary albumin-to-creatinine ratio, systolic blood pressure, cardiovascular disease and diabetes.

†The top 3 proteins, ordered by absolute coefficient weight, were RNASE1 (pancreatic ribonuclease), IGFBP4 (insulin-like growth factor-binding protein 4), and GDF15 (growth differentiation factor 15).

risk stratification and targeted prevention in a high-risk hypertensive population.

Importantly, subgroup analyses stratified by follow-up duration revealed that model performance was maintained—and numerically improved—for events occurring beyond the first tertile compared with those within the shortest tertile. This temporal pattern, observed consistently across both cohorts despite differences in their respective tertile cutoffs, suggests that the proteomic risk score captures stable, medium- to long-term susceptibility to CKD in hypertensive patients, rather than merely reflecting preclinical disease states that precede imminent diagnosis. The sustained predictive performance over ≈6 to 8 years supports the score's potential utility for risk stratification in clinical practice, where a 5- to 10-year prediction window is often most relevant for guiding preventive interventions.

Notably, the 3-protein score (RNASE1, IGFBP4, GDF15) achieved 98% to 99% of the full score's predictive performance. These proteins reflect key mechanisms in kidney impairment. GDF15 is linked to inflammation and fibrosis[31]; IGFBP4 modulates metabolic and vascular pathways[32]; and RNASE1 is implicated in angiogenesis and vascular injury.[33] The parsimony of this 3-protein panel enhances its potential for translation into a clinically accessible assay. However, prospective studies are needed to demonstrate its utility for risk stratification and to determine whether its use can guide targeted interventions and improve outcomes in high-risk hypertensive patients.

Limited Utility of the eGFR-Based PRS in This Context

In contrast, the eGFR-based PRS provided negligible improvement when added to the clinical model. This aligns with evidence that PRSs offer limited incremental value in older adults, where environmental exposures and acquired risk factors dominate disease risk.[34] In established hypertension, the proteome appears to capture active disease pathways more effectively than static genetic predisposition.

Target Name	Method	nSNP	BETA	SE	P value
BMPER	Inverse variance weighted	11	0.341035292	0.151856446	0.024718473
GFRA1	Inverse variance weighted	7	0.356269875	0.153085766	0.019951607
CST3	Inverse variance weighted	6	0.363630778	0.139860942	0.009323866
CXCL16	Inverse variance weighted	6	0.59254975	0.223506052	0.008021642
FOLR1	Inverse variance weighted	4	0.597893553	0.254902873	0.018997806
AMBP	Inverse variance weighted	3	0.798177805	0.278379975	0.004140895
STC1	Inverse variance weighted	2	1.246138089	0.616349188	0.043196464

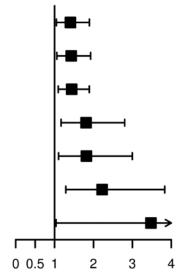

Figure 2. Mendelian randomization identifies 7 proteins associated with hypertensive chronic kidney disease (CKD).
Forest plot showing Mendelian randomization estimates for 7 proteins with evidence of association with hypertensive CKD ($P<0.05$). Squares represent effect estimates (β coefficients); horizontal lines indicate 95% CIs. Number of genetic variants used as instruments (nSNP) is shown for each protein. Interpretation: Positive β values indicate that genetically predicted higher protein levels are associated with increased CKD risk, suggesting potential causal relationships. All 7 proteins showed effect directions consistent with observational Cox regression analyses. SNP indicates single nucleotide polymorphism.

Unraveling Causal Pathways and Therapeutic Potential

Moving beyond prediction, our hypothesis-generating MR analysis prioritized 7 proteins with potential relevance to hypertensive CKD, mitigating concerns of confounding and reverse causation inherent in observational studies. These proteins fall into 3 categories. First, 2 proteins offer novel insights. BMPER and GFRA1, primarily known for roles in endothelial function, vascular development, and kidney organogenesis, were unexpectedly identified, opening new avenues for investigating developmental programming in adult kidney disease under hypertensive stress.[35,36] Second, several proteins reinforce established pathways. CST3 is a well-established marker of diabetic nephropathy[37]; CXCL16 and AMBP align with renal inflammation and protease inhibition, respectively.[38,39] Third, FOLR1 is critical for cellular folate uptake, and STC1 is linked to phosphate metabolism and cellular stress, pointing to less conventional mechanisms beyond traditional hemodynamic and inflammatory models.[40,41] Notably, established CKD markers such as TCEA2 (transcription elongation factor A protein 2), NRG4 (neuregulin-4), and GCKR (glucokinase regulatory protein)[42] were not identified in our causal analysis, suggesting they may have limited causal relevance specifically in hypertensive kidney disease. This diversity of implicated pathways underscores the etiologic heterogeneity of CKD and suggests that proteomic risk models may need to be tailored to specific populations or disease subtypes.

Enrichment analysis consolidated these findings into 2 key themes: dysregulated protease inhibition and perturbations in folate metabolism, the latter driven by FOLR1. Druggability assessment highlighted FOLR1 and STC1 as therapeutically relevant, though not yet explored in CKD. FOLR1, which mediates renal tubular folate uptake, is targeted by approved antagonists (mirvetuximab soravtansine for ovarian cancer; pyrimethamine for malaria) and 10 investigational agents.[43,44] The directionality of our MR finding aligns with an antagonist strategy, such that higher FOLR1 levels increase risk. However, the approved agents' toxicity profiles, including increased serum creatinine with mirvetuximab soravtansine, and their oncological or antiparasitic indications preclude direct repurposing for CKD. While this safety signal may be confounded by the underlying conditions, it underscores the need for caution. Developing novel FOLR1 inhibitors optimized for renal safety represents a logical next step. More broadly, this example highlights the challenge of interpreting pQTL MR for circulating proteins, as the relevant causal tissue may not be blood.

Despite the biological plausibility of these findings, they should be interpreted with caution. Colocalization support for shared causal variants with CKD was limited (PH4<0.5 for all), and none of the MR associations survived false discovery rate correction. These proteins should therefore be considered prioritized candidates requiring functional validation rather than confirmed causal mediators.

Clinical Implications

These findings have several clinical implications. First, the protein risk score, particularly the 3-protein panel, could be developed into a clinical test to identify high-risk hypertensive patients for intensive monitoring and early intervention (eg, stricter blood pressure control, sodium-glucose cotransporter 2 inhibitors). Second, the study shifts the paradigm from traditional risk factors to a pathway-based understanding of CKD risk in hypertension. Third, the integrated MR-druggability pipeline nominates specific proteins, especially FOLR1, for further investigation as therapeutic targets.

Limitations

Several limitations should be acknowledged. First, the predominantly European ancestry limits generalizability to other ethnic groups.[45] Second, the modest number of events in the external validation cohort (n=72) limited statistical power for subgroup analyses and may have contributed to variability in performance metrics. Validation in larger-scale, more diverse external cohorts is needed. Third, only ≈50% of participants met the hypertension definition based on a prior medical diagnosis or antihypertensive medication use; the remainder were classified solely by elevated blood pressure at baseline. However, subgroup analyses showed consistent model performance regardless of how hypertension was defined. Fourth, the Olink Explore platform used in this study, while comprehensive, covers ≈3000 proteins and may not capture all potentially relevant biomarkers for CKD progression. Future studies using broader proteomic platforms or complementary assays could identify additional predictive proteins. Fifth, the MR analysis should be interpreted as hypothesis-generating, prioritizing 7 proteins for functional validation rather than providing definitive causal evidence. This exploratory approach, combined with limited colocalization support, underscores the need for experimental confirmation of their roles and mechanisms in hypertensive kidney disease.

Perspectives

Our study demonstrates that plasma proteomics significantly enhance the prediction of incident CKD in hypertension beyond clinical and genetic models. A compact 3-protein signature (RNASE1, IGFBP4, GDF15) captures most of this predictive utility. MR analysis nominates 7 proteins with potential causal roles, implicating pathways of protease inhibition and folate metabolism. Among these, FOLR1 and STC1 emerge as druggable targets. These findings illustrate the power of integrating multimodal data to advance from risk prediction to causal understanding and therapeutic innovation in hypertensive kidney disease.

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Author Contributions

Y. Huang and X. Qin designed and conducted the research. Y. Huang, D. Chen, and Z. Ye performed the data management and statistical analyses. Y. Huang and X. Qin wrote the article. All authors reviewed/edited the article for important intellectual content. All authors read and approved the final article.

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Disclosures

None.

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